



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Main)

UNIT TEST

04-07-2021

JEE(Main + Advance) : Enthusiast Course (PHASE : S,I(A),I, PHASE-I LIVE & PHASE

Test Type : UNIT TEST

Test Pattern : JEE (Main)

TEST DATE : 04-07-2021

ANSWER KEY

PHYSICS

SEC-A	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	C	A	C	B	B	D	C	D	A
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	C	C	B	A	C	D	C	B	D	D
SEC-B	Q.	1	2	3	4	5	6	7	8	9	10
	A.	5.6	7.00	2.50	0.87	2.00	13.00	12.50	1.41 to 1.42	4.00	12.00

CHEMISTRY

SEC-A	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	D	C	B	B	A	C	C	D	D
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	D	B	D	D	D	C	B	D	C
SEC-B	Q.	1	2	3	4	5	6	7	8	9	10
	A.	80.00	1.33	4.00	38.00	37.50	5.00	9.10 to 9.40	4.00	3.00	5.00

MATHEMATICS

SEC-A	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	A	B	B	C	D	C	C	B	B
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	D	C	A	C	A	D	A	C	D
SEC-B	Q.	1	2	3	4	5	6	7	8	9	10
	A.	2.00	3.00	0.00	27.00	2.00	0.00	4.00	20.78	0.00	0.00

HINT – SHEET

PHYSICS

SEC-A

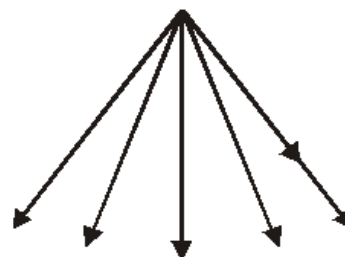
1. Ans (A)

$$a = \frac{F}{m} = \frac{GMm}{\frac{R^3}{m}} = \frac{GM}{R^3}r = \frac{g}{R}r$$

$$\omega = \sqrt{\frac{g}{R}}$$

$$v_{\max} = A\omega = \frac{R}{2} \sqrt{\frac{g}{R}} = \frac{\sqrt{gR}}{2}$$

2. Ans (C)



$$F = \frac{GMm}{a^2} + \frac{GMm}{4a^2} + \frac{GMm}{3a^2} \times \sqrt{3}$$

$$\frac{Gm^2}{a^2} \left\{ \frac{5}{4} + \frac{1}{\sqrt{3}} \right\}$$

3. **Ans (A)**

Escape velocity is independent of mass and location

4. **Ans (C)**

$$R = \frac{\rho \ell^2}{A \ell}$$

$$\frac{R'}{R} = \left(\frac{\ell'}{\ell} \right)^2 = \left(\frac{2+0.2}{2} \right)^2 = (1.1)^2$$

$$= 1.21$$

$$\frac{R' - R}{R} = 0.21$$

5. **Ans (B)**

$$N = \int_0^5 t dt + 6 \int_0^5 t^2 dt$$

$$= \frac{4}{2} \times 25 + \frac{6}{3} \times 125$$

$$= 50 + 250 = 300 \text{ C}$$

6. **Ans (B)**

$$\frac{k\lambda}{L\sqrt{3}} 2 \sin 30 = \frac{kQ}{2L^2\sqrt{3}}$$

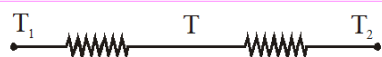
7. **Ans (D)**

Formule

8. **Ans (C)**

Properties of conductor

9. **Ans (D)**



$$\frac{T - T_1}{R_1} + \frac{T - T_2}{R_2} = 0$$

$$T_1 R_2 + T_2 R_1 = T(R_1 + R_2)$$

10. **Ans (A)**

Material for which $\Delta \ell$ less strip bond towards that end.

11. **Ans (C)**

$$\rho = \frac{m}{v} = \frac{m}{\ell^3}$$

$$\frac{d\rho}{d} = 3 \frac{d\ell}{\ell}$$

$$= 3 \times 0.01$$

$$\rho\% \text{ change} = 0.03\%$$

12. **Ans (C)**

$$Q_g = m \times 540$$

$$Q_T = 1 \times 80 + 1 \times 100 = 180$$

$$Q_g = Q_T$$

$$m \times 540 = 180$$

$$m = \frac{1}{3}$$

$$m_{\text{steam}} = 1 - \frac{1}{3} = \frac{2}{3}$$

14. **Ans (A)**

$$k = \frac{Y A}{L}$$

$$f = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

15. **Ans (C)**

$$\hat{r} = \hat{e} - 2(\hat{n} \cdot \hat{e}) \cdot \hat{n}$$

16. **Ans (D)**

$$A_1 \omega_1 = A_2 \omega_2$$

$$A_1 \sqrt{\frac{k_1}{m}} = A_2 \sqrt{\frac{k_2}{m}}$$

17. **Ans (C)**

$$\frac{\sigma}{\epsilon_0} = \frac{2k\lambda}{r} = \frac{2}{4\pi\epsilon_0} \frac{\lambda}{r}$$

$$\sigma = \frac{1}{2\pi} \frac{\lambda}{r} = \frac{8}{6 \times 3.14}$$

$$= 0.42$$

18. **Ans (B)**

Theory

19. **Ans (D)**

$$H = \frac{KA \Delta T}{\Delta x}$$

$$ML^2 T^{-3} = (K) \frac{L^2 K}{L}$$

$$ML^2 T^{-3} K^{-1} = [K]$$

20. **Ans (D)**

$$T = \frac{2\pi}{2\omega} = \frac{\pi}{\omega}$$

PHYSICS

SEC-B

1. **Ans (5.6)**

$$i = \frac{6-4}{10} = \frac{1}{5}$$

$$v_y - \frac{8}{5} - 4 = v_x$$

$$v_y - v_x = 5.6 \text{ v}$$

2. **Ans (7.00)**

$$\frac{v_1}{v_2} = \frac{10}{\left(\frac{1000}{70}\right)} = \frac{7}{10}$$

$$v_1 + v_2 = 17$$

$$\begin{cases} v_1 = 7v \\ v_2 = 10v \end{cases}$$

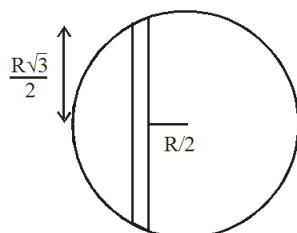
3. **Ans (2.50)**

$$\begin{aligned} P &= h\rho g \\ &= 5 \times 10^3 \times 10^3 \times 10 \\ &= 5 \times 10^7 \end{aligned}$$

$$\frac{\Delta v}{v} = 2 \times 10^{-2}$$

$$B = \frac{P}{\frac{\Delta v}{v}} = \frac{5 \times 10^7}{2 \times 10^{-7}} = 2.5 \times 10^9$$

4. **Ans (0.87)**



$$N = \frac{4\pi G \rho \frac{R}{2} m}{3}$$

$$N_{fr} = \frac{\mu 4\pi G \rho R m}{6} \times \frac{R\sqrt{3}}{2}$$

S.H.M.

$$k_i = 0$$

$$k_f = \frac{1}{2} m \left(\frac{R\sqrt{3}}{2} \right)^2 \left\{ \frac{4\pi G \rho}{3} \right\}$$

WET

$$w_{fr} = k_f - 0$$

$$\mu = \frac{\sqrt{3}}{2}$$

5. **Ans (2.00)**

$$B = \frac{P}{\frac{\Delta v}{v}}$$

$$\frac{\Delta v}{v} = \frac{P}{B}$$

$$\frac{3\Delta r}{r} = \frac{P}{B} = \frac{6 \times 10^9}{10 \times 10^{10}}$$

$$\frac{\Delta r}{r} = \frac{2}{100}$$

6. **Ans (13.00)**

$$\frac{5}{8} \times 2\pi = \frac{5\pi}{4}$$

$$t = \frac{5\pi T}{4 \times 2\pi} = \frac{5T}{8}$$

7. **Ans (12.50)**

$$Q = i^2 R T$$

$$10 \times 10^{-3} = (2 \times 10^{-3})^2 \times R \times 2$$

8. **Ans (1.41 to 1.42)**

$$i = \frac{3}{4} \times 60 = 45$$

$$\delta m = 2i - A = 30$$

$$\mu = \frac{\sin\left(\frac{A+\delta m}{2}\right)}{\sin\left(\frac{A}{2}\right)} = \frac{\frac{1}{\sqrt{2}}}{\frac{1}{2}} = \sqrt{2}$$

9. **Ans (4.00)**

$$2 = 2\pi \sqrt{\frac{\ell}{g}}$$

$$g = 2\pi^2$$

10. **Ans (12.00)**

$$Q_{net} = 0 = Q_1 - Q_2$$

$$0 = \omega_1 \delta_1 - \omega_2 \delta_2$$

$$0.02 \times (1.5 - DA_1) = 0.03 (1.6 - DA_2)$$

$$A_1 = 1.8 A_2$$

$$\text{Now } \delta_{net} = 2^\circ = (\mu_1 - 1)A_1 - (\mu_2 - 1)A_2$$

$$= \left(\frac{1}{2}\right) A_1 - (0.6) \frac{A_1}{1.8}$$

$$2 = \frac{A_1}{6}$$

$$A_1 = 12$$

CHEMISTRY

SEC-A

8. **Ans (C)**

$$\frac{-dp}{dt} = K P_{NO}^a P_{H_2}^b$$

For 1st experiment, hydrogen pressure is

$$\text{constant } \frac{-dp}{dt} = K^1 P_{NO}^a$$

$$2.25 \text{ torr/sec} = k^1 \alpha (450 \text{ torr})^a \text{ ---(I)}$$

$$0.25 \text{ torr/sec} = k^1 \alpha (150 \text{ torr})^a \text{ ---(II)}$$

$$\text{On dividing } \frac{2.25}{0.25} = \left(\frac{450}{150} \right)^a = 3^a$$

$$9 = 3^a \Rightarrow a = 2$$

For IInd experiment, NO pressure is constant

$$1.90 = k^{ii} (290)^b \text{ ---(III)}$$

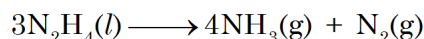
$$0.951 = k^{ii} (145)^b \text{ ---(IV)}$$

$$\text{On Diving } \frac{1.90}{0.95} = \left(\frac{290}{145} \right)^b$$

$$2 = 2^b \Rightarrow b = 1$$

$$\boxed{\frac{-dp}{dt} = K P_{NO}^2 P_{H_2}}$$

10. **Ans (D)**



$$W = -\Delta n_g RT = -\frac{5}{3} \times 8.3 \times 300 = -4150 \text{ J}$$

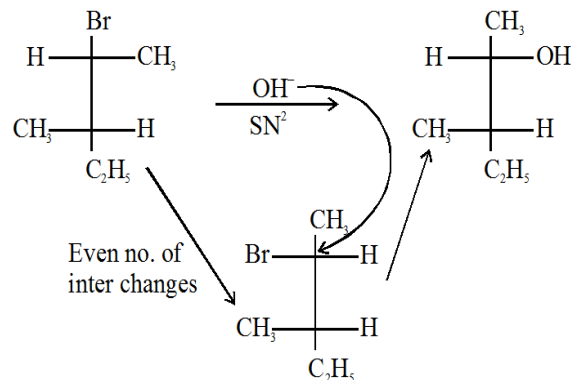
11. **Ans (B)**

$$A_t = \frac{A_0}{2^3} = \frac{10^7}{8} \text{ dps}$$

$$\Rightarrow = 2 A_t =$$

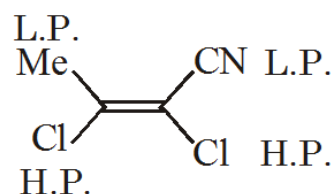
$$= 2.5 \times 10^6 \text{ s}^{-1}$$

13. **Ans (B)**



Option B is correct

17. **Ans (C)**



High priority group same side "Z" form.

CHEMISTRY

SEC-B

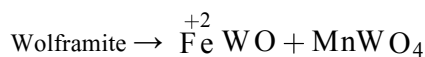
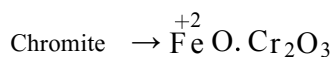
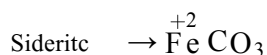
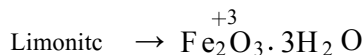
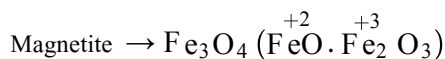
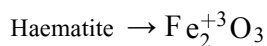
1. **Ans (80.00)**

$$\Delta_t = \frac{4}{9} \Delta_0 = \frac{4}{9} \times 18000$$

$$x = 8000$$

$$\Rightarrow \frac{x}{100} = 80$$

2. **Ans (1.33)**



$$\frac{x}{y} = \frac{4}{3} = 1.33$$

5. **Ans (37.50)**

let mmoles of each is = x

n-factor of FeO = 1

n-factor of Fe_{0.80}O = 0.4

m_{eq} of FeO + m_{eq} of Fe_{0.80}O = m_{eq} of KMnO₄

$$x \times 1 + x \times 0.4 = 35 \times 0.3 \times 5$$

$$x = 37.50 \text{ mmoles}$$

$$\text{mmoles of Fe}^{3+} \text{ produced} = 37.5 + 37.5 \times 0.8 = 67.50 \text{ mmoles}$$

6. **Ans (5.00)**

5

7. **Ans (9.10 to 9.40)**

$$z = 1 + \frac{Pb}{RT}$$

$$\frac{dz}{dp} = \frac{1}{10} = \frac{L}{RT}$$

$$b = \frac{RT}{10} = \frac{22.4}{10} = 2.24$$

$$4 \times V \times N_0 = 2.24$$

$$V = \frac{2.24}{4 \times 6} \times 10^{-23} \ell$$

$$= \frac{2.24 \times 1000}{24} \times 10^{-23} \text{ cm}^3$$

$$= 9.33 \times 10^{-22} \text{ cm}^3$$

9. **Ans (3.00)**

a, c, e

MATHEMATICS

SEC-A

1. **Ans (B)**

$$\text{Put } \begin{vmatrix} (1-\lambda) & 3 & -4 \\ 1 & -(3+\lambda) & 5 \\ 3 & 1 & -\lambda \end{vmatrix} = 0$$

$$\Rightarrow \lambda(\lambda+1)^2 = 0 \Rightarrow \lambda = -1, 0$$

2. **Ans (A)**

$$\frac{\cos A}{2R \sin A} = \frac{\cos B}{2R \sin B} = \frac{\cos C}{2R \sin C}$$

$$\cot A = \cot B = \cot C \quad \therefore A = B = C$$

Hence triangle is equilateral

$$\text{Hence area} = \frac{\sqrt{3}}{4} (a)^2 = \sqrt{3}$$

3. **Ans (B)**

$$f'(x) = 8x + 24x^3 + 48x^5$$

and

$$f''(x) = 8 + 72x^2 + 240x^4 \geq 0 \text{ for all } x \in \mathbb{R}$$

$$\therefore f'(x) = 0 \Rightarrow x(8 + 24x^2 + 48x^4) = 0$$

$$\Rightarrow x = 0 \quad \text{Since } f''(0) > 0$$

$\therefore f$ has a local minima at $x = 0$.

There is no other critical point and $f'(x)$

exists for all $x \in \mathbb{R}$

4. **Ans (B)**

$$f'(x) = 3x^2 - 15x + 18$$

$$= 3(x^2 - 5x + 6)$$

$$\begin{array}{c} f'(x) > 0 \quad f'(x) < 0 \quad f'(x) > 0 \\ \hline \quad \quad \quad 2 \quad \quad \quad 3 \end{array}$$

$$x^2 - 4x \leq 0 \Rightarrow 0 \leq x \leq 4$$

$$\begin{array}{c} f(x) \\ \text{Increases} \quad \text{Decreases} \quad \text{Increases} \\ \hline 0 \quad \quad 2 \quad \quad 3 \quad \quad 4 \end{array}$$

$$f(z) = 8 - 30 + 36 + 2020$$

$$= 2034$$

$$f(4) = 64 - 120 + 72 + 2020 = 2036$$

5. **Ans (C)**

$$\text{Given } \frac{ds}{dt} = 16$$

$$\frac{d}{dt} (4\pi r^2) = 16$$

$$4\pi 2r \frac{dr}{dt} = 16$$

$$\frac{dr}{dt} = \frac{2}{\pi r} \quad \dots (1)$$

$$\text{Now } \frac{dv}{dt} = \frac{d}{dt} \left(\frac{4}{3} \pi r^3 \right)$$

$$= \frac{4}{3} \pi 3r^2 \frac{dr}{dt} = 4\pi r^2 \frac{2}{\pi r} = 8r$$

6. **Ans (D)**

Do Yourself.

7. **Ans (C)**

Do Yourself.

8. **Ans (C)**

Use graph

OR

Let $0 \leq \alpha, \beta, \gamma, \delta \leq \pi$

such that $\sec \alpha = 3, \sec \beta = 4, \sec \gamma = -3, \sec \delta = -4$

then we can say $\alpha < \beta < \delta < \gamma$

9. **Ans (B)**

$$\int \frac{x^7}{(2x^2 + 1)^5} dx = \int \frac{x^{-3}}{(2 + x^{-2})^5} dx$$

$$\text{Let } x^{-2} = t \Rightarrow x^{-3} dx = -\frac{1}{2} dt$$

$$= -\frac{1}{2} \int \frac{dt}{(2+t)^5} = \frac{1}{8} \frac{x^8}{(2x^2 + 1)^4} + C$$

10. **Ans (B)**

$$\int e^x (-\cot x + \operatorname{cosec}^2 x) dx$$

$$= -e^x \cot x + c$$

11. **Ans (A)**

Do Yourself.

12. **Ans (D)**

Apply LMVT

$$f(x) = \frac{f(5) - f(-2)}{5 - (-2)} \text{ for some } x \text{ in } (-2, 5)$$

$$\text{Now, } -4 \leq \frac{f(5) - f(-2)}{7} \leq 3$$

$$-28 \leq f(5) - f(-2) \leq 21$$

$\therefore$ Greatest possible value of $f(5) - f(-2)$ is 21.

13. **Ans (C)**

$$f(x) = \begin{cases} x^2 - ax + 3, & x \text{ is rational} \\ 2 - x, & x \text{ is irrational} \end{cases}$$

is continuous when

$$x^2 - ax + 3 = 2 - x \text{ or } x^2 - (a-1)x + 1 = 0$$

which must have two distinct roots for which (a

$$-1)^2 - 4 > 0$$

$$\Rightarrow (a-1-2)(a-1+2) > 0$$

14. **Ans (A)**

$$f(x) = 1 + x^3$$

$$\int (1 + x^3) dx = x + \frac{x^4}{4} + C$$

15. **Ans (C)**

$$\sin A + \sin B = \frac{c(a+b)}{c^2} = \frac{a+b}{c} =$$

$$\frac{k \sin A + k \sin B}{k \sin C}$$

Thus $C = \pi/2$

$\Rightarrow$ ABC is right angled.

Hence (C) is the correct answer.

16. **Ans (A)**

$$I = \int \left(\frac{1 - \tan^2 x}{1 + \tan^2 x} \right) \frac{1}{\sin 2x} dx = \int \frac{\cos 2x}{\sin 2x} dx$$

$$= \int \cot 2x dx = \frac{\ln |\sin 2x|}{2} + C$$

17. **Ans (D)**

$$\text{limit} = \frac{f'(1)}{2} = \frac{\pi - 2}{16\sqrt{2}}$$

18. **Ans (A)**

$$\int e^{\sin x} \left(\frac{5 + \cos^2 x}{(2 - \sin x) \sqrt{3 + \cos^2 x}} \right) \cos x dx$$

put $\sin x = t$

$$\int e^t \left(\frac{6 - t^2}{(2 - t) \sqrt{4 - t^2}} \right) dt$$

$$\int e^t \left(\frac{4 - t^2}{(2 - t) \sqrt{4 - t^2}} + \frac{2}{(2 - t) \sqrt{4 - t^2}} \right) dt$$

$$\int e^t \left(\sqrt{\frac{2+t}{2-t}} + \frac{2}{(2-t)^{3/2}(2+t)^{1/2}} \right) dt$$

$$\text{If } g(t) = \sqrt{\frac{2+t}{2-t}}, g'(t) = \frac{2}{(2-t)^{3/2}(2+t)^{1/2}}$$

$$e^t \sqrt{\frac{2+t}{2-t}} + c$$

$$f(x) = e^{\sin x} \sqrt{\frac{2 + \sin x}{2 - \sin x}} + c$$

$$f(0) = 1 \Rightarrow c = 0$$

$$f(x) = e^{\sin x} \sqrt{\frac{2 + \sin x}{2 - \sin x}}$$

$$f\left(\frac{\pi}{2}\right) = \sqrt{3}e$$

19. **Ans (C)**

$$y' = \frac{8x}{3(4 - y^2)} \Rightarrow x = 0$$

$$\Rightarrow y = 0, \pm 2\sqrt{3}$$

20. Ans (D)

$$I = \int \frac{\sqrt{x-1}}{x\sqrt{x+1}} dx = \int \frac{x-1}{x\sqrt{x^2-1}} dx$$

$$= \int \frac{dx}{\sqrt{x^2-1}} - \int \frac{dx}{x\sqrt{x^2-1}}$$

$$= \ln|x + \sqrt{x^2+1}| - \sec^{-1}x + c$$

MATHEMATICS

SEC-B

1. Ans (2.00)

$$f(x) = x^3 + x$$

$$f'(x) = 3x^2 + 1 \Rightarrow f(x) \text{ is } \uparrow$$

$$f(f(x)) > f(4-2x) \Rightarrow f(x) > 4-2x$$

$$\Rightarrow x^3 + x > 4-2x \Rightarrow x^3 + 3x - 4 > 0$$

$$\Rightarrow x^3 - x + 4x - 4 > 0$$

$$\Rightarrow x(x^2 - 1) + 4(x - 1) > 0$$

$$\Rightarrow (x-1)(x^2 + x + 4) > 0 \Rightarrow x > 1$$

Least integral value of x is 2.

2. Ans (3.00)

$$f(-1) = 4$$

$$f(-1) = k + 1$$

$$k + 1 \leq 4$$

$$\Rightarrow k \leq 3$$

3. Ans (0.00)

$$f(x) + f(y) = f(x)f(y) + f(xy)$$

partial differentiation wrt x

$$f'(x) = f'(x)f(y) + f'(xy) \cdot (y)$$

put x = 1

$$-2 = -2f(y) + f'(y)y$$

$$\text{get } f(y) = 1 - y^2 \Rightarrow f(x) = 1 - x^2$$

Now, number of solution(s) of $f(x) = 2$

$$1 - x^2 = 2 \Rightarrow x^2 = -1 \Rightarrow x \in \phi$$

4. Ans (27.00)

$$C_1 \rightarrow C_1 - bC_3, C_2 \rightarrow C_2 + aC_3$$

$$\Delta = \begin{vmatrix} 1+a^2-b^2+2b^2 & 0 & -2b \\ 0 & 1-a^2+b^2+2a^2 & 2a \\ 2b-b+a^2b+b^3 & -2a+a-a^3-ab^2 & 1-a^2-b^2 \end{vmatrix}$$

$$= \begin{vmatrix} 1+a^2+b^2 & 0 & -2b \\ 0 & 1+a^2+b^2 & 2a \\ b(b^2+a^2+1) & -a(1+a^2+b^2) & 1-a^2-b^2 \end{vmatrix}$$

$$= (1+a^2+b^2)^2 \begin{vmatrix} 1 & 0 & -2b \\ 0 & 1 & 2a \\ b & -a & 1-a^2-b^2 \end{vmatrix}$$

$$= (1+a^2+b^2)^2 ((1-a^2-b^2) + 2a^2 - 2b(-b))$$

$$= (1+a^2+b^2)^3$$

Now, given that $ab = 1$

$$\Rightarrow a^2 + b^2 \geq 2ab \Rightarrow a^2 + b^2 \geq 2$$

So, minimum value is $3^3 = 27$

5. Ans (2.00)

$$T_n = \cot^{-1} \left(n^2 + \frac{3}{4} \right)$$

$$= \tan^{-1} \left(\frac{1}{n^2 + (3/4)} \right)$$

$$= \tan^{-1} \left(\frac{1}{1+n^2 - (1/4)} \right)$$

$$= \tan^{-1} \left(\frac{1}{1 + \left(n - \frac{1}{2} \right) \left(n + \frac{1}{2} \right)} \right)$$

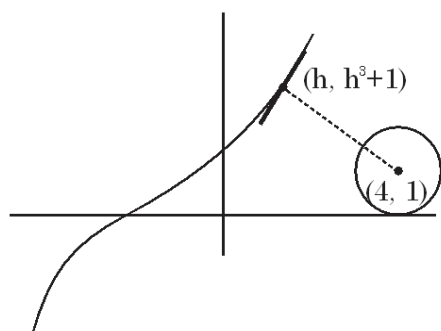
$$= \tan^{-1} \left(\frac{\left(n + \frac{1}{2} \right) - \left(n - \frac{1}{2} \right)}{1 + \left(n + \frac{1}{2} \right) \left(n - \frac{1}{2} \right)} \right)$$

6. Ans (0.00)

$$\begin{aligned} \text{Using } \int_0^1 f(x) dx &= \int_0^1 f(1-x) dx \\ I &= \int_0^1 \sqrt[3]{x^2(2x-3) + (1-x)} dx \\ &= \int_0^1 \sqrt[3]{(1-x)^2(-1-2x) + x} dx \\ &= - \int_0^1 \sqrt[3]{(x^2-2x+1)(1+2x) - x} dx \\ &= - \int_0^1 \sqrt[3]{2x^3-3x^2-x+1} dx = -I \\ 2I &= 0 \quad \therefore I = 0 \end{aligned}$$

7. Ans (4.00)

$$y = x^3 + 1$$



$$y' = 3x^2$$

$$y' \big|_{(h, h^3+1)} = 3h^2$$

$$\therefore -\frac{1}{3h^2} = \frac{h^3 + 1 - 1}{h - 4}$$

$$-(h-4) = 3h^5$$

$$\boxed{3h^5 + h = 4}$$

8. Ans (20.78)

$$27\Delta^2 = s^4$$

$$27s(s-a)(s-b)(s-c) = s^4$$

$$3[(s-a)(s-b)(s-c)]^{1/3} = s$$

$$\frac{s-a+s-b+s-c}{3} \geq [(s-a)(s-b)(s-c)]^{1/3}$$

$$\therefore s-a = s-b = s-c$$

$$\therefore a = b = c$$

$$\therefore a = 2R \sin A = 2 \times 4 \times \frac{\sqrt{3}}{2} = 4\sqrt{3}$$

$$\text{Area} = \frac{\sqrt{3}}{4} (4\sqrt{3})^2 = 12\sqrt{3}$$

9. Ans (0.00)

$$I = \int_{1/5}^5 \frac{1}{x} 2^{x^2 + \frac{1}{x^2}} \left(x^3 - \frac{1}{x^3} \right) dx$$

$$\text{let } x = \frac{1}{t} \Rightarrow dx = \frac{-1}{t^2} dt$$

$$I = \int_5^{1/5} t 2^{t^2 + \frac{1}{t^2}} \left(t^3 - \frac{1}{t^3} \right) \cdot \frac{1}{t^2} dt$$

$$I = - \int_{1/5}^5 \frac{1}{t} 2^{t^2 + \frac{1}{t^2}} \left(t^3 - \frac{1}{t^3} \right) dt$$

$$I = -I \Rightarrow 2I = 0 \Rightarrow I = 0$$

10. Ans (0.00)

$$f'(x) = 4x^3 + 18x^2 + 70x + 6$$

$$\text{and } f''(x) = 12x^2 + 36x + 70 > 0 \quad \forall x \in \mathbb{R}$$

so $f(x)$ has no point of inflection.